

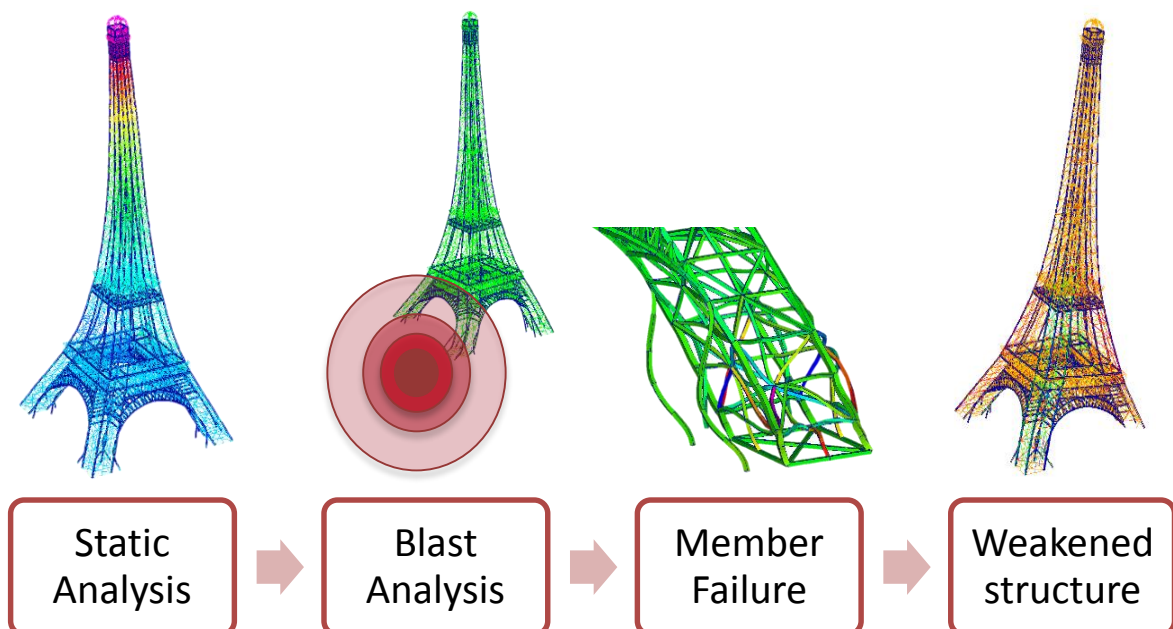
Blasting Response of the Eiffel Tower

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Abstract. A finite element model of the Eiffel Tower was constructed using Strand7 software. The model replicates the existing tower, with dimensions justified through the use of original design drawings. A static and dynamic analysis was conducted to determine the actions of the tower under permanent, imposed and wind loadings, as well as under blast pressure loads and earthquake loads due to an explosion. It was observed that the tower utilises the full axial capacity of individual members by acting as a ‘truss of trusses’. As such, permanent and imposed loads are efficiently transferred to the primary columns through compression, while wind loads induce tensile forces in the windward legs and compressive forces in the leeward. Under blast loading, the tower experienced both ground vibrations and blast pressures. Ground vibrations induced a negligibly small earthquake loading into the structure which was ignored in subsequent analyses. The blast pressure was significant, and a dynamic analysis of this revealed that further research is required into the damping qualities of the structure due to soil and mechanical properties. In the worst case scenario, the blast was assumed to completely destroy several members in the adjacent leg. Despite this weakened condition, it was observed that the tower would still be able to sustain static loads, at least for enough time for occupant evacuation. Further, an optimised design revealed the structure was structurally sound under a 46% reduction of the metal tower’s mass.



INTRODUCTION

The Eiffel Tower was constructed as the leading monument of the Universal Exposition to mark the centenary of the French Revolution of 1789. The large iron archways served as the gateway to Champ-de-Mars, with the Tower itself acting as the key attraction of the Universal Exposition. It was the first structure to reach the coveted 1000 feet, and maintained its status as the world's tallest structure until 1930. While today the Eiffel Tower is held in great revere, this was not always the case with calls for the Tower to be demolished in the early 20th century. The height of the Tower proved to be its saving grace with city officials recognising its significance as a radio tower, which proved instrumental in later years during WWI and WWII.

The Eiffel Tower is one of the world's most iconic monuments, with the architectural wonder attracting more visitors than any other paid tourist attraction in the world. In addition to its architectural value, the tower also represents significant improvements in engineering design with the four legs representing a roughly exponential curve to best resist wind loads (Weidman and Pinelis, 2004), allowing it to reach heights never before attained by any previous structure. Since the tower was initially constructed as a temporary structure, the focus of the following analysis is to both determine the adequacy of the structure to resist permanent loads, and the extent to which the tower was over designed with respect to its capacity to resist permanent and imposed loads. For the purpose of this analysis, loads will be based on Australian Standards to ensure rigour, rather than the original hand-calculated loads applied to the structure during its design. The European Standards were not established at the time of construction, and thus have not been included. An analysis has also been conducted to determine the capacity of the tower to resist blast loads resulting from a nearby explosion.

The Eiffel Tower is a large wrought iron lattice tower, comprising 7,300 tonnes of iron and 2.5 million rivets. The three observation decks, which are serviced by multiple stairways and elevators, attract an average of 25,000 visitors each day. This makes the observation decks of the tower some of the most highly foot trafficked areas in the world. The steel decking is required to resist significant imposed actions associated with this foot traffic. Additionally, the tower contains two restaurants that offer panoramic views of the surrounding city for diners.

TABLE 1. Tower Details

Location: Paris, France	Year Construction Complete: 1889
Architect: Stephen Sauvestre	Approximate Cost: 7,800,000 gold francs (equivalent current-day AU\$43 million)
Structural Engineers: Maurice Koechlin, Emile Nouguier	Overall Height: 324m
Function: Originally an entrance to the 1889 World's Fair. Then used for radio and television transmission. Now operates as a tourist attraction.	Floor Area: approximately 6200 m ²
The Structure of the Plan: Wrought iron lattice	
Number of Floors: 3 floors above ground	

STRUCTURAL MEMBERS

Floor system: Wrought iron deck supported underneath by a grid of large trusses that span between the legs.

Beams: Wrought iron trusses that span between the four legs, providing both load continuity and platform decking support. The lower and middle platforms are supported by a 16x16 and 6x6 roughly square grid with an area of 4200m² and 1650m² respectively. Beams in the upper half of the tower provide continuity between the four legs before they join to form a single leg. The spans of these beams vary from 1.8m to 15.7m.

Legs: The tower consists of four separate curved legs that meet to form a single point at the peak. Each of these legs are approximately 15m by 15m and consist of four columns that are connected by an iron lattice, allowing the columns to behave like a truss. The four columns contained in each of the legs is assembled from a series of angles and flat sections that form a roughly square hollow cross section (see Figure 1). The diagonal bracing connecting the four columns to form a single leg consists of trusses. The section properties of both the square column sections and the diagonal bracing members vary with height, reflecting the increase in permanent loads of the tower at lower heights. For the purpose of finite element modelling, the trusses and square hollow columns were represented as single beam elements with appropriate section properties given to match the section properties of the actual truss system / column section. An example calculation is provided below for a column section at ground level.

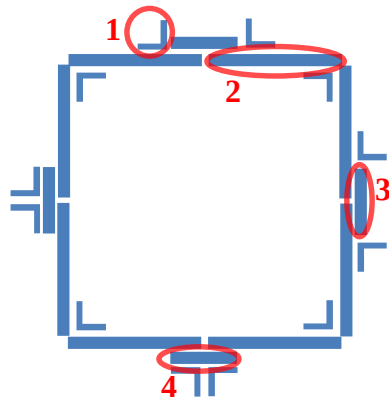


TABLE 2. Structural Properties for Section 1

Member	Dimensions	Area	$I_{x/y}$ ($\text{mm}^4 \times 10^3$)	J ($\text{mm}^4 \times 10^3$)
1	100 x 100 x 12	2256	2520.5	108.3
2	800 x 18.5	14800	422.1	1688.4
3	200 x 12	2400	28.8	115.2
4	205 x 12	2460	29.5	118.1

$$\begin{aligned}
 \text{Total area} &= \Sigma \text{Area} = 95992 \text{ mm}^2 & (1) \\
 \text{Total } I_x / I_y &= \Sigma I_{\text{own axis}} + \Sigma A(y_o - \bar{y})^2 = 10.12 \times 10^9 \text{ mm}^4 & (2) \\
 \text{Total } J &= \Sigma J = 8.52 \times 10^6 \text{ mm}^4 & (3)
 \end{aligned}$$

FIGURE 1. Section Properties for Section 1

Foundations: Each of the four legs rest on masonry blocks and are anchored into the foundation below the blocks by a total of sixteen anchors that have a diameter of 10cm and a length of 7.5m.

Ground Conditions: The soil conditions consist of both hard and soft soil. The soft soil is predominately located underneath the riverside legs, requiring deeper and larger foundations than the park side legs to reduce settlement of the tower.

TABLE 3. Structural Elements for Section 1 of the Tower

Details of the Structural Elements	Available Structural Element Sizes	Suggested Structural Element Sizes
Columns - Structural Systems	Uniform Box sections	Respective Second moment of areas inputted as detailed above
Beams - Structural Systems	T sections	T-sections varying from 60x120x6mm to 180x360x20mm
	Angle sections	Angle-section 60x60x6mm
Grated Floor System	Quad 4 Plates	20mm thick membrane

STRUCTURAL SYSTEM

The structural system of the Eiffel Tower grants the ability to withstand significant vertical loads from both the weight of the tower itself and imposed loading from the heavy foot traffic associated with visitors. The main components that resist vertical loads are each of the four columns located at the corners of all four support legs (16 in total). The size of these roughly square hollow sections varies according to height, with larger section dimensions located at the base of the tower. A series of diagonal truss elements join each of the four columns together to form a leg, while horizontal trusses also span between each of the columns at regular intervals. These members effectively brace the columns, reducing their effective lengths, which subsequently increases their resistance to global buckling.

Due to the heavy foot traffic associated with the large number of tourists packed onto the observation decks, imposed loads experienced by the platform decking are significant. The wrought iron decking on the lower platform is supported on the underside by large trusses that transfer the load to the four legs. Additionally, the arches serve to transfer some load from the platform supporting trusses to the legs of the tower. Due to the curvature of the tower, the second platform has a significantly smaller area. The platform requires fewer supporting trusses and is instead split into a sixteen metre, roughly square grid. The top platform only consists of an overhang around the perimeter of the lattice leg support. The overhang is supported by diagonal columns protruding from the tower a small distance below the platform such that it forms an approximately 30-degree angle with the tower. Overhangs of the lower two platforms are supported by diagonal columns which are connected to the legs where possible and are otherwise connected to the bottom chord of the truss supporting the platform.

Samples of the design used to model the Eiffel Tower are provided (see Figure 2, Figure 3 and Figure 4).

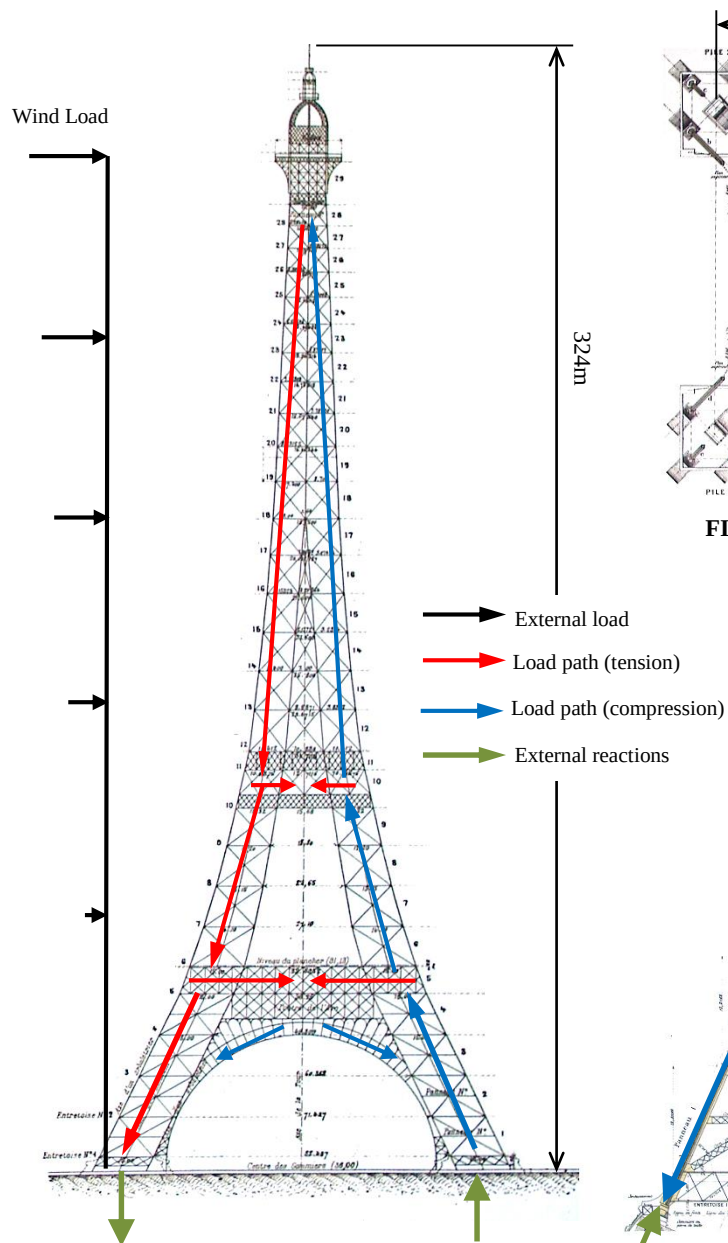


FIGURE 3. Elevation View Eiffel Tower (wind load paths)

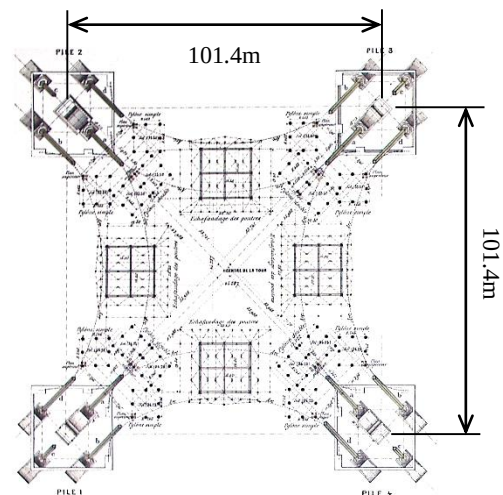


FIGURE 2. Plan View Eiffel Tower

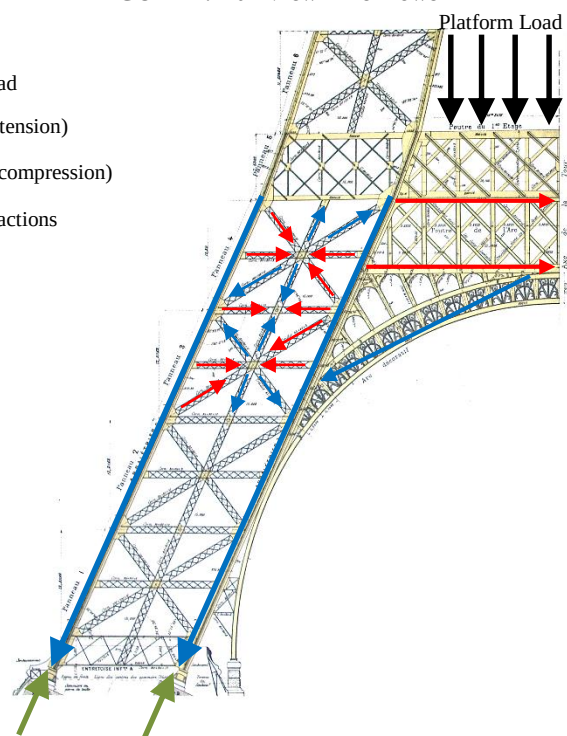


FIGURE 4. Permanent / imposed load paths

Lateral loads on the structure are predominately associated with wind loadings on the side of the tower. The curvature of the four supporting legs provides a base area that is significantly larger than the area at the top of the tower. This increases the tower's resistance to overturning moments caused by the wind, since the restoring force has an increased moment arm (see Figure 3). Similarly, by narrowing as the tower climbs in height the tower presents a reduced surface area for the wind to act on which proves significant because the wind pressure and overturning moment arm both increase with height. To ensure that lateral load is transferred and the four legs do not act independently from one another, continuity between the four legs is provided at the lower and intermediate platforms by large trusses that span between the legs. Concurrently, smaller horizontal trusses spanning between the legs provide continuity for the top half of the tower, prior to merging of the four legs into a single leg.

LOADS

Permanent Loads: The weight of the tower was applied to the structure through the use of a gravity load case. This resulted in a weight approximately equal to the quoted weight for the metal structure, 7,300 tonnes (Lemoine, 2008). While this accounted for the structural members of the Eiffel Tower, it failed to account for additional permanent loads from items such as internal stair wells and elevator shafts. This was accounted for by increasing

the density of the member components of the supporting legs. The density was increased in accordance with a sum of reaction forces at the base equal to the weight of the whole structure, 10,100 tonnes (Lemoine, 2008).

Imposed Loads: Areas of the tower that experience imposed loading are limited to the bottom, intermediate, and top platforms. Imposed loads result from the presence of restaurants and the heavy foot traffic of the observation decks. As a consequence, the imposed loading on these areas was conservatively assumed as 7.5kPa in accordance with AS1170.1 for areas susceptible to overcrowding.

Wind Loads: The wind loading varied as a function of height and was assumed to apply only to the exposed faces of the beam and plate members on one side of the tower. The width of the exposed area of the beams was taken conservatively as 800mm for all beams located below the middle platform, and 600mm for all beams located above it. Wind loading was calculated in accordance with AS1170.2 (equation 4), and verified against fluid mechanics derivations (equation 5). Wind speed varies for an external flow due to viscosity and drag effects.

$$p = 0.5 \rho_{air} [V_{des,\theta}]^2 C_{fig} C_{dyn} \quad (4)$$

$$\frac{U}{u^*} = 2.44 \cdot \ln \frac{\delta u^*}{\nu} + 5 \quad (5)$$

Where δ is the height of the boundary layer, U is the wind speed at a height δ , u^* is the wind speed variable, and ν is the viscosity. The wind speed, u^* and $V_{des,\theta}$ above, was observed to vary logarithmically with height. A cubic function was derived to approximate this logarithmic variation to enable it being entered into Strand7. This returned an equation for pressure, which was multiplied by the respective width, depending at what height the pressure was applied. The resulting equation was input as a linearly varying distributed load.

The four legs of the tower act as trusses themselves to resist permanent and imposed actions. Diagonal members experience tensile and compressive forces in an alternating fashion, while horizontal and vertical members are under tension and compression, respectively (see Figure 4). These members impart loads on the main columns of the structure, which take on the bulk of the compressive stress. The trusses supporting the first, second, and third platforms, act to not only distribute the imposed loading at these levels, but also to redistribute any overlying loading (permanent and imposed) to the members below. The aim of this is to more evenly distribute the load to underlying members. As observed in a static analysis of the Strand7 model under permanent and imposed loading (see figure 5), member stresses above the second platform truss are fairly uniform. Below the truss, stresses are reorganised such that alternating diagonal members experience tension and compression, similar to a truss. Increased permanent loading is experienced by members further down the tower, due to self-weight, so this reorganisation is lessened until the next platform truss redistributes the loading. The load paths associated with resistance to lateral loading can be thought of more simplistically due to the geometry of the tower. The splayed base provides a significant restoring moment, due to the increased lever arm, such that the windward legs experience tension and the leeward legs experience compression (see Figure 6).

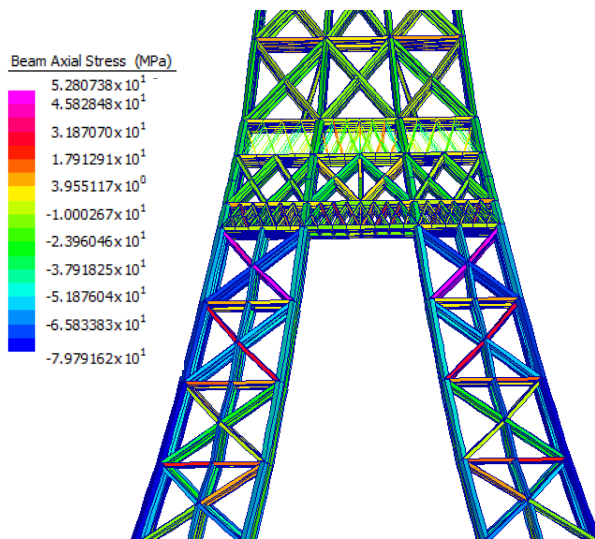


FIGURE 5. Load paths in model under 1.2G + 1.5Q

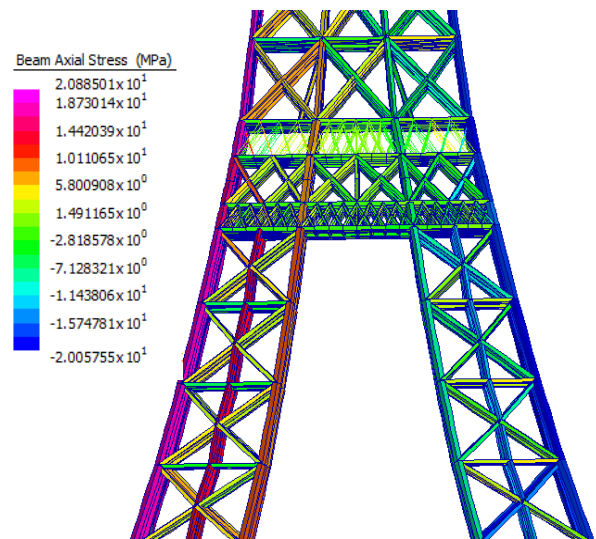


FIGURE 6. Load paths in model under wind loading

As shown in Figure 3 previously, a restoring couple is expected at the base of the structure. From the finite element model, it is observed that the maximum moment is at the fixed node on the base of the innermost primary column on the leeward side (-689kNm), while the minimum moment is similarly at the fixed node on the base of the innermost primary column on the windward side (639kNm). This indeed creates a significant restoring couple to resist lateral loading. Maximum bending stress is experienced by diagonal members directly beneath the redistribution trusses under the second platform, for both load combinations 1.2G + 1.5Q (52.8MPa) and 1.2G + W + ϕ_c Q (41.4MPa). Similarly, minimum bending stress is experienced by primary column members at the base for both load combinations, -162.2MPa and -132.8MPa respectively.

Blast Loads: As a result of the explosion, two different load components were analysed. The magnitude of the explosion was taken as the equivalent of 21kg of TNT, since reliable data was obtained from the 2013 fertilizer plant explosion in Texas, USA (Grinberg and Yan, 2016). The two load components resulting from the explosion comprised of air pressure waves and ground vibrations. The blast pressure varied based on its distance from the focal point of the explosion and was calculated using equations 6, 7, and 8 provided below (Lam, Mendis, & Ngo, 2004; Ngo, Mendis, Gupta, & Ramsay, 2007). The focal point was determined one metre from one of the tower's legs in an attempt to balance reasonability and worst case. The ground vibrations caused by the explosion were equivalent to a magnitude 2.1 earthquake on the Richter scale (Grinberg and Yan, 2016). Consequently, table 6.4 from the AS1170.4 standards was used to model the effect of this size earthquake on the soft soil conditions that the tower rests on. The relevant ground acceleration used for the model was defined according to a regression analysis by Wald et al. (1999) to give a value of 0.0017g. Since the explosion occurred on the surface, critical ground accelerations were assumed to occur only in the x and y axes.

$$Z = \frac{R}{W^{1/3}} \quad (6)$$

$$P_{so} = \frac{6.7}{Z^3} + 1 \text{ bar } (P_{so} > 10 \text{ bar}) \quad (7)$$

$$P_{so} = \frac{0.975}{Z} + \frac{1.455}{Z^2} + \frac{5.85}{Z^3} - 0.019 \text{ bar } (0.1 \text{ bar} < P_{so} < 10 \text{ bar}) \quad (8)$$

TABLE 4. Blast Loading with Distance

R: Distance from blast (m)	Z: Distance vs weight TNT Factor (m/kg^{1/3})	P_{so}: Blast pressure (kPa)
1	0.36246	141.7
5	1.812301	1.944789
10	3.624601	0.483595
20	7.249202	0.158541
30	10.8738	0.087521
40	14.4984	0.05709
50	18.12301	0.040212

In addition to variation with radial distance from the focal point, the blast pressure also varied with time (see Figure 7). An amplification factor of 1.8 due to the blast pressure waves reflecting upwards off the ground was also included in the model. A peak blast loading was used for subsequent static loading, however a dynamic analysis was required to observe the reaction of the tower as the blast load decreased with time.

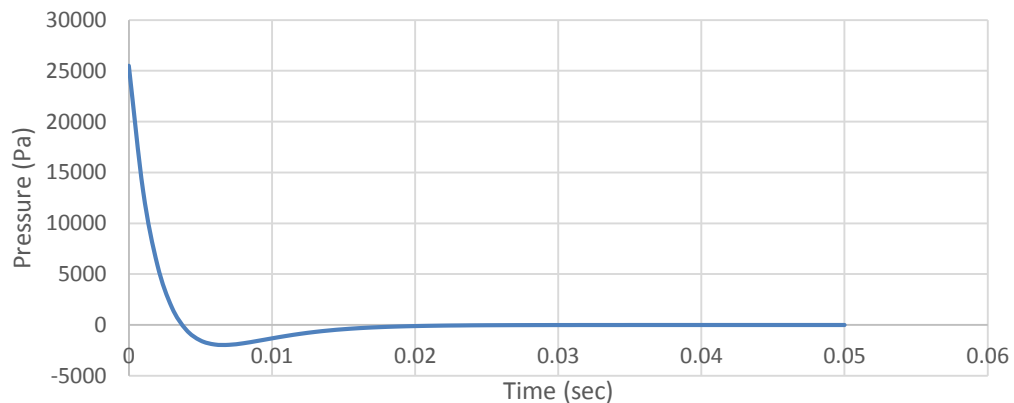


FIGURE 7. Blast pressure vs time

NUMERICAL ANALYSIS

The finite element model of the Eiffel Tower was constructed from beam2 and quad4 elements that matched the properties of the members used for construction of the tower. The Young's Modulus (E) and Poisson's Ratio (ν) of the wrought iron were modelled as 172 000 MPa and 0.28 respectively. Restraint was provided at the base of all four legs with fully fixed restraint applied to model the effect of bolting to the sizeable foundations. The entire lattice structure was created by first modelling a quarter of the structure, and mirroring this along the tower's two axes of symmetry. This quarter of the structure was manually created using dimensions from the original construction drawings found in Lemoine's *Gustave Eiffel: The Eiffel Tower* (2008). Once mirrored, quad4 elements were placed at the three platform heights primarily to facilitate the application of an imposed pressure. A lack of information regarding the current flooring within these platforms meant the plates have a thickness that approximates the expected permanent loads due to flooring. They also provide some structural support to surrounding beams.

Linear Static Analysis: The effect of permanent, imposed and wind loads on the structure, which are outlined in the previous section, were evaluated using a linear static analysis. The following load combination cases were applied in accordance with AS1170.0;

$$1.2G + 1.5Q \quad (9)$$

$$1.2G + W + \phi_c Q \quad (10)$$

Where G represents permanent loading, Q represents imposed loading, W represents wind loading and ϕ_c represents the worst-case distributed imposed load combination factor, equal to 0.6. The maximum vertical displacement and axial stress resulted from the $1.2G + 1.5Q$ load combination, the results of which are displayed in Figure 8 and Figure 9 respectively. As expected, the maximum vertical deflection of the tower was small, measuring 91mm at the peak of the tower. Maximum axial compressive stress was experienced at the base of the innermost columns at 162MPa. Comparing this to the expected tensile strength of wrought iron in Europe towards the end of the 19th century, 358MPa (Kelton, Arwade & Lutenegeger, 2011), the tower has an estimated safety factor of 2.27 based on strength.

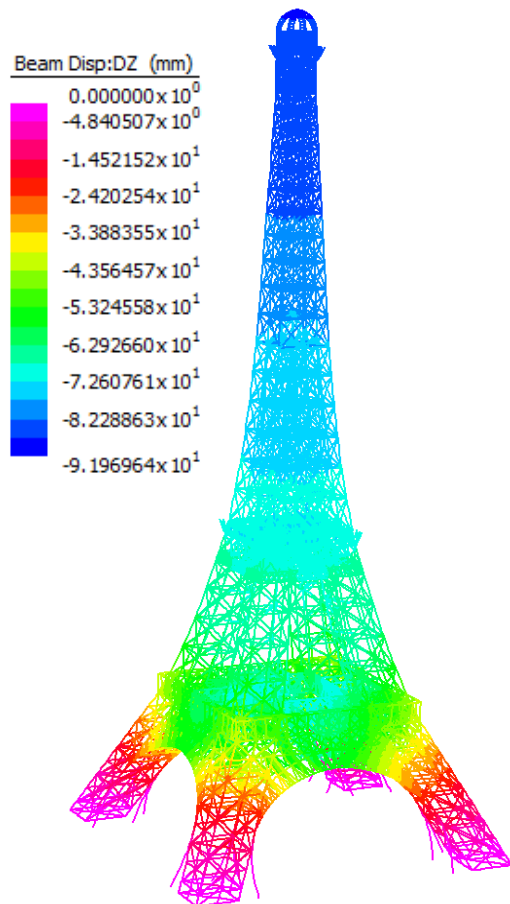


FIGURE 8. Vertical displacement under load combination $1.2G + 1.5Q$

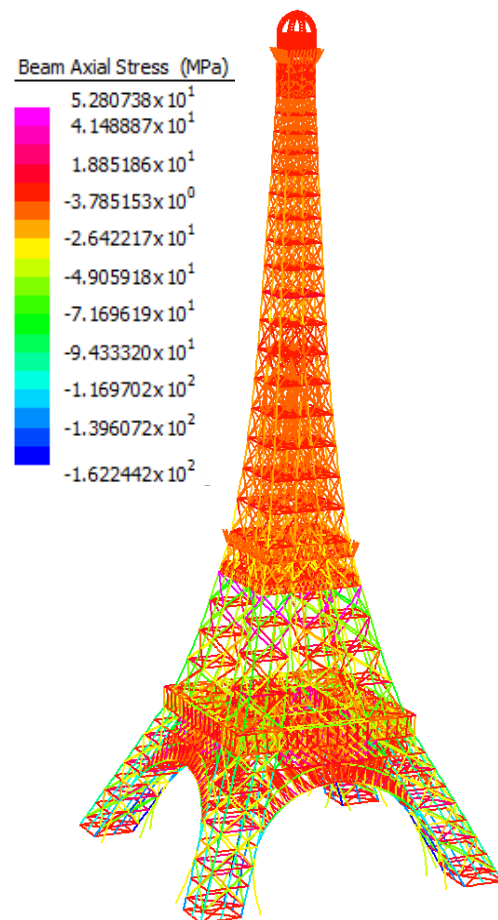


FIGURE 9. Axial stress under load combination $1.2G + 1.5Q$

Conversely, maximum lateral displacement was present under the load combination of $1.2G+W+\phi_c Q$, the results of which are displayed in Figure 10. Maximal lateral displacement is observed to be 245mm at the peak of the tower. Serviceability limit state criteria from AS1170.0 suggests a maximum lateral displacement of height/500, or 565mm. Comparing the two, the tower has an estimated safety factor of 2.31 based on serviceability. Figure 11 displays the axial stress throughout the tower for the load combination of $1.2G+W+\phi_c Q$. Maximum tensile and compressive stresses are observed to be lower than the load combination of $1.2G + 1.5Q$.

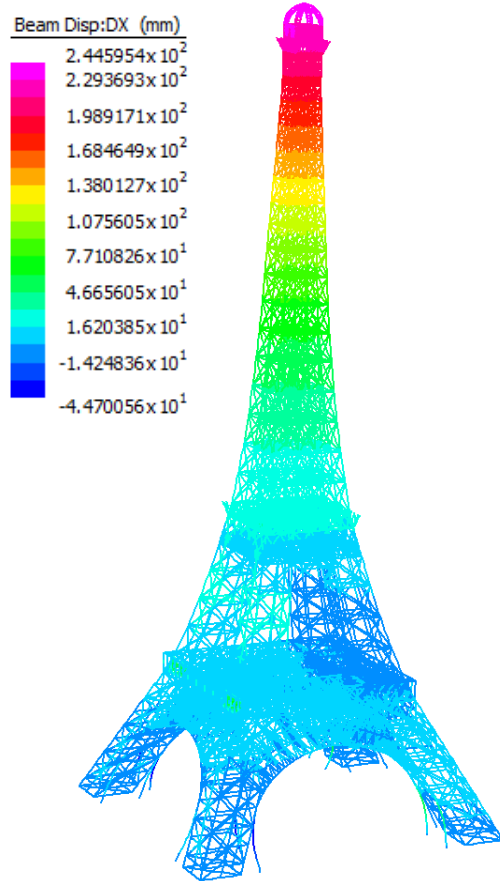


FIGURE 10. Lateral displacement under load combination $1.2G + W + \phi_c Q$

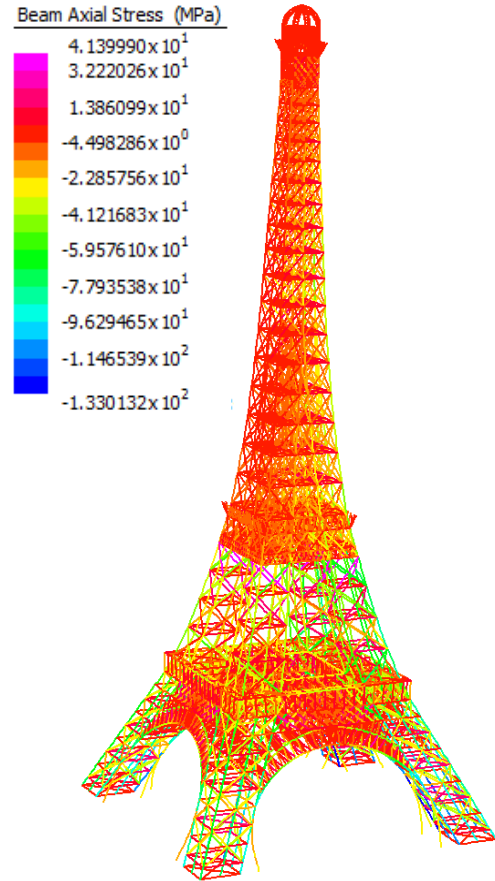


FIGURE 11. Axial stress under load combination $1.2G + W + \phi_c Q$

Blast Analysis: The effect of the ground vibrations due to the blast load were determined by first performing a natural frequency analysis of the tower. The first 6 modes that converged were then used, in conjunction with the period vs time graph (AS1170.4) to perform the spectral response analysis. Ground accelerations were based on historical data of earthquakes with similar magnitudes.

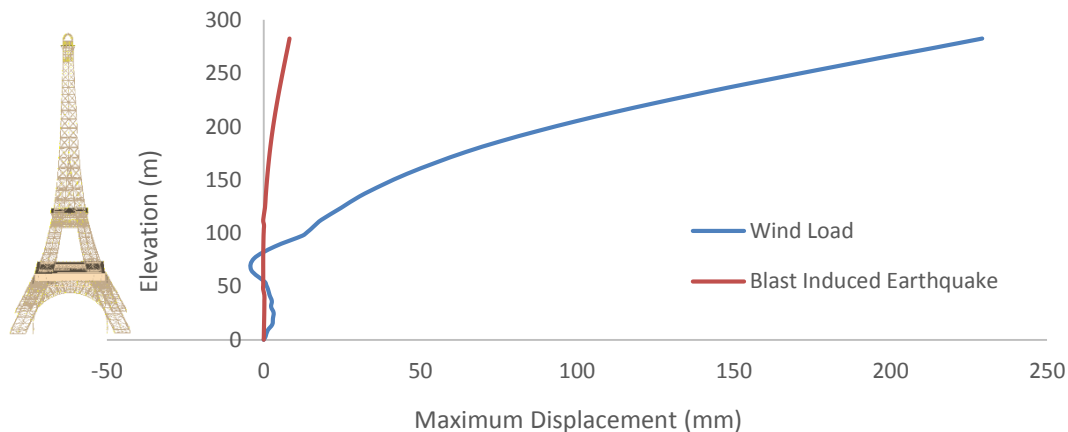


FIGURE 12. Maximum horizontal displacement with elevation

Figure 12 graphically represents the lateral displacement of the tower with its height due to wind and earthquake loading. The lateral displacement due to ground vibrations from the blast are seen to be insignificant in comparison to that experienced during wind loading. Instead, blast pressure is the critical load caused by the explosion. The effect of the blast pressure as a function of time was examined using a non-linear transient dynamic analysis, where factors with time were derived from Figure 7. A non-linear static analysis was conducted to determine the initial conditions for the dynamic analysis, which was imported into the solver. These results were compared with the linear static results, with no significant difference between the two.

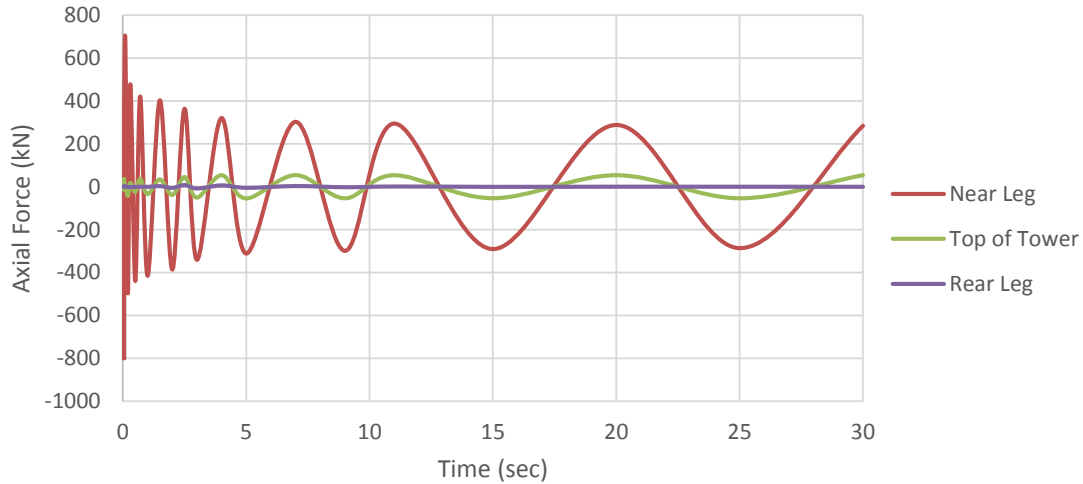


FIGURE 13. Blast induced axial force over time

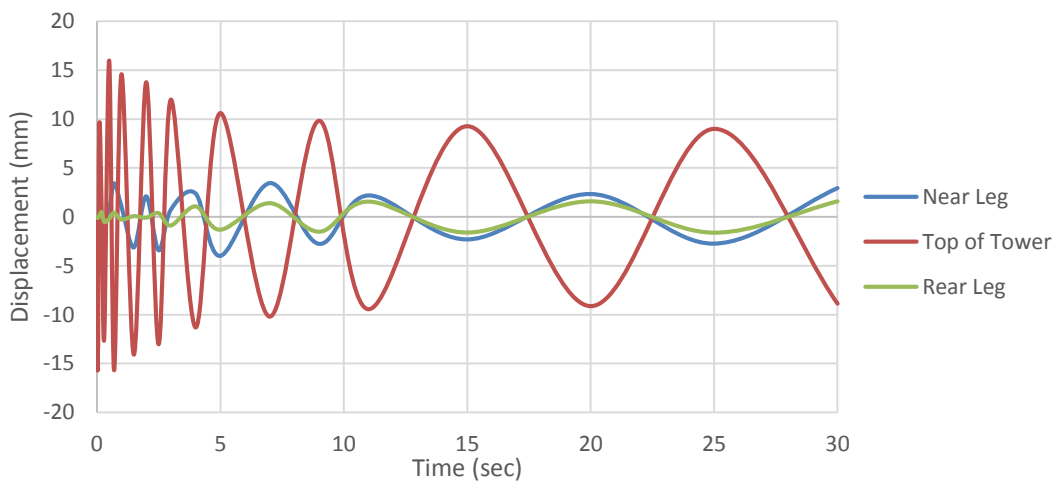


FIGURE 14. Blast induced displacement over time

The blast pressure induces large stresses in the leg closest to the blast, which propagates through the structure (see Figure 16) to cause oscillating positive and negative stresses in the members (see Figure 17). This reverberation is large within the time period analysed, however the effects of damping have not been taken into account and as such, beyond the first oscillations (within the first few seconds), the effects of inertia are observed to dominate. This is seen in Figure 13 and Figure 14, where over a longer period of time, the period of the oscillations increases while the amplitude remains constant. Inertial forces multiply the effects of the reverberation while in reality, damping due to soil and material properties is expected. The oscillation, provided all members remain intact, is estimated to reduce drastically with time as the stresses are distributed throughout the structure and into the soil underneath. Damping properties of the structure and soil have not been analysed within this finite element model, and should be subject to future analysis. There is, however, a substantial time lag between the first induced member stresses close to the blast, and those in members further from the blast (see Figure 15), which are also orders of magnitude smaller. It seems more likely that the members in the leg adjacent to the blast will exceed yield stress, and the stresses elsewhere in the structure will dissipate through damping. Thus the damping qualities are no longer of interest, but rather the structural capabilities of the weakened structure after the blast has caused damage to the structure.

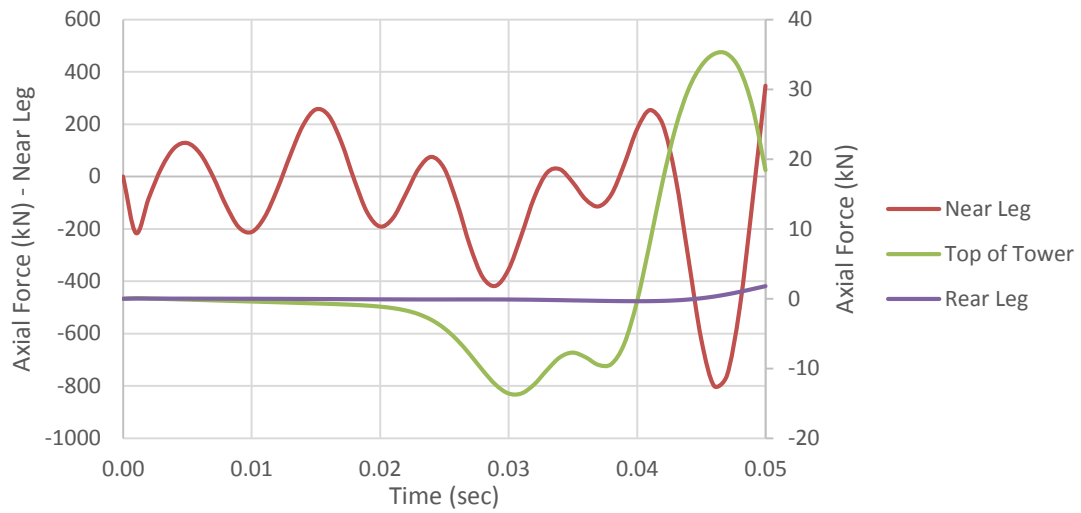


FIGURE 15. Time lag of force across tower

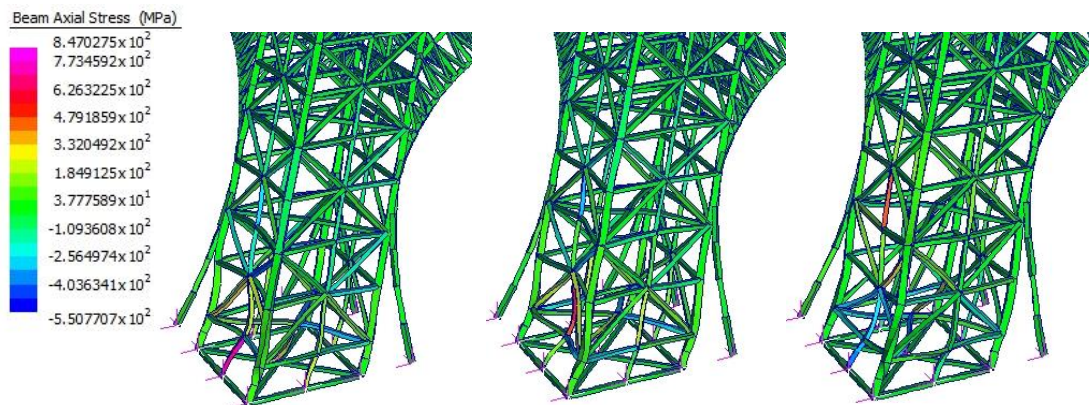


FIGURE 16. Time-phased propagation of axial stress close to blast at times 0.005, 0.007, and 0.013 s

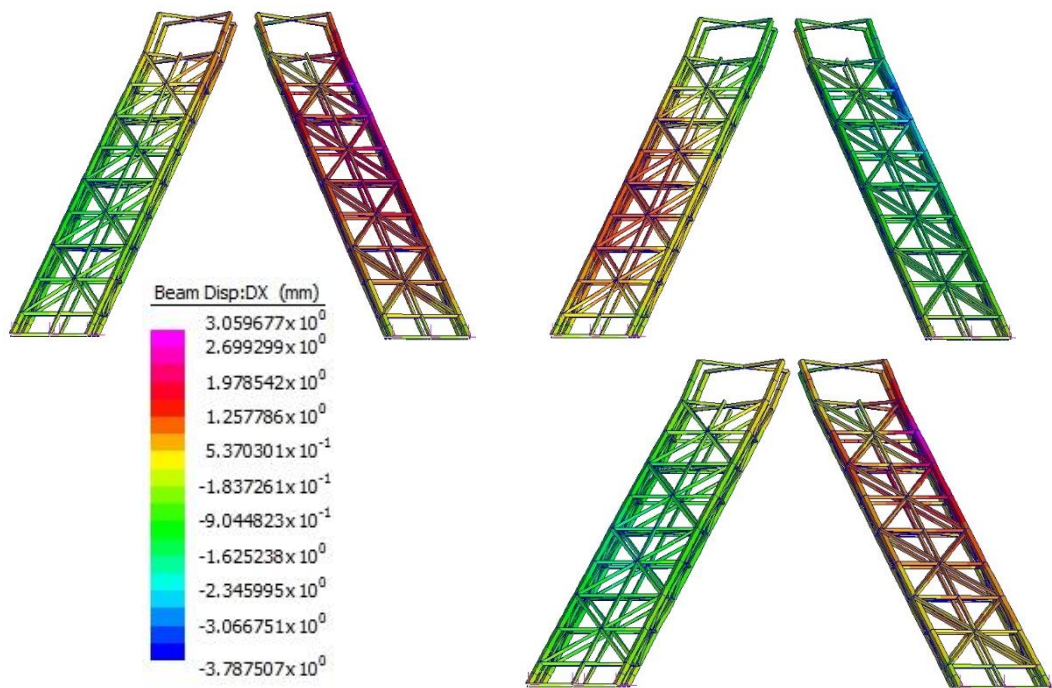


FIGURE 17. Oscillation in rear legs at 3 seconds (top-left), 4 seconds (top-right), and 5 seconds (bottom-right) from time of explosion

Elements with an axial stress that exceeds the ultimate stress of wrought iron, 358MPa (Kelton, Arwade & Lutenege, 2011), were removed from the model through an iterative process whereby members affected by the blast load were removed and again the structure reanalysed. The subsequent weakened structure was analysed under static loading. The recoil due to the blast pressure was not critical in comparison to static loading, and thus a static analysis sufficiently analysed the weakened structure to determine its structural integrity after the explosion.

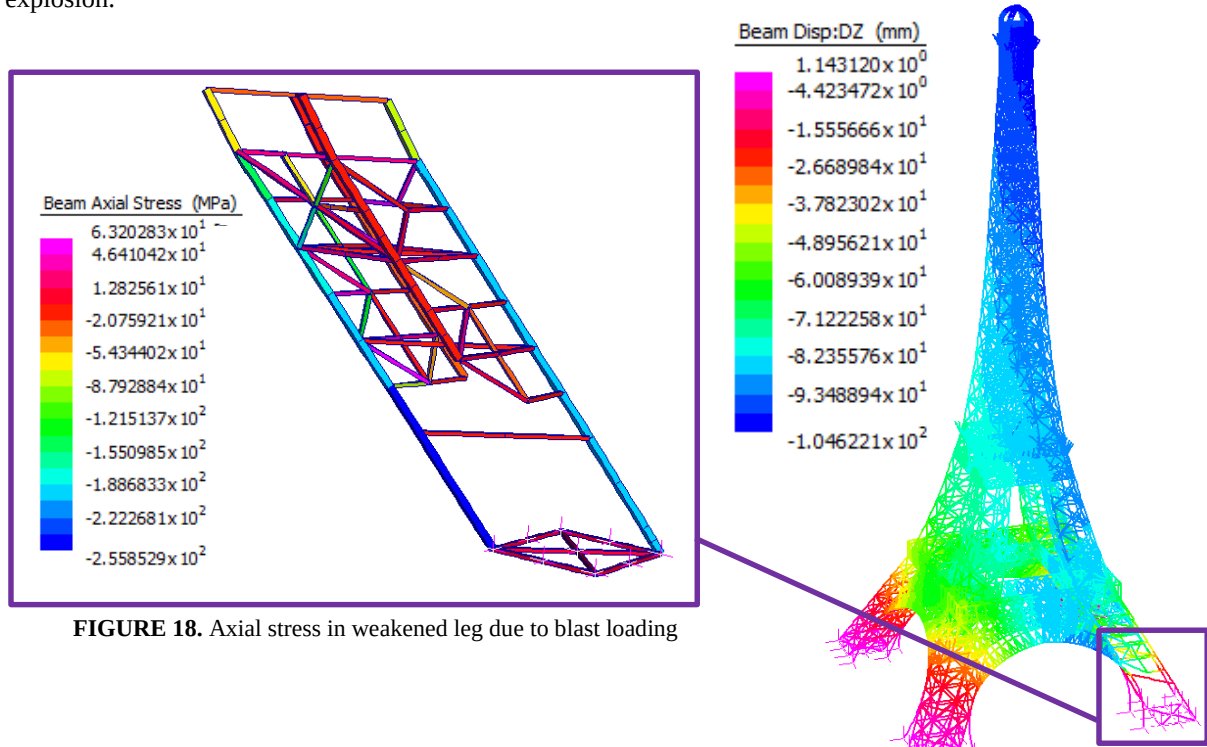


FIGURE 18. Axial stress in weakened leg due to blast loading

FIGURE 19. Static analysis of weakened structure

The maximum axial stress was experienced in the interior primary column within the affected leg. This is expected as the lean of the deformed structure places large compressive stresses on this member. It is possible that the remaining columns of the damaged leg could undergo buckling since the support provided from bracing is removed, increasing their effective lengths. It should be noted that despite these large stresses, the remaining three legs experience tension less than the tensile capacity of the member and thus the tower is expected to remain standing under static loading, at least for some period of time. It is also noteworthy that when completely removing this leg, the tower still remains structurally sound, indicating that regardless of whether buckling occurs or not the structure maintains its resistance to toppling. This is a result of the large footings, which allow the tower to hang off the leg furthest from the affected leg, placing this in tension.

STRUCTURAL DESIGN

The finite element model uses a variety of member types that aim to replicate existing beams used to construct the Eiffel Tower. Where possible, member properties have been calculated from these existing beams and substituted into the model. An example of this is seen in the Structural Members section of this report.

All beams are assumed to be purely under compression or tension due to the truss-like nature of the tower. Despite applying to steel members, AS4100-1998 has been used to determine sectional and member capacities for the wrought iron beams. The following analysis is for the primary column section at ground level, which approximates the cross-section of the beam as a square hollow section. The section and member capacities were determined for the members that contributed most to the weight of the tower (see table 5).

$$N^* \leq \phi N_s \quad (11)$$

$$\phi N_s = \phi k_f A_g f_y = 0.9 \times 1.0 \times 95992 \times 207 = 17883 \text{ kN} \quad (12)$$

$$\text{where } b_e = b \left(\frac{\lambda_{ey}}{\lambda_e} \right) = 850 \cdot \left(\frac{40}{27.4} \right) \leq b \quad (13)$$

$$N_c = \alpha_c N_s \leq N_s \quad (14)$$

$$\lambda_n = \left(\frac{l_e}{r} \right) \left(\sqrt{k_f} \right) \sqrt{\frac{f_y}{250}} = \left(\frac{7500}{324.7} \right) (\sqrt{1}) \sqrt{\frac{207}{250}} = 21.02 \quad (15)$$

$$\alpha_c = 0.979 \quad (16)$$

$$\phi N_c = 17650 \text{ kN} \quad (17)$$

$$\text{Maximum axial stress} = \frac{\phi N_c}{A_g} = \frac{17650}{95992} = 183.83 \text{ MPa} \quad (18)$$

TABLE 5. Existing member capacities

Member	Existing Members	Load Capacity (MPa)	Max Axial Stress (MPa)	Length of Section (m)
Bracing Section I	EA 100 x 100 x 11	186.30	141.67	6804.28
Columns Section I	SHS 850 x 850 x 28.3	183.87	162.24	1058.69
Bracing Section II	EA 80 x 80 x 9	182.05	76.43	4026.06
Columns Section II	SHS 800 x 800 x 21.2	176.84	81.94	978.96
Bracing Section III	EA 90 x 90 x 12	179.72	31.82	6909.34
Columns Section III	SHS 600 x 600 x 27.5	170.17	44.28	1187.06
Bracing Section IV	EA 90 x 90 x 11	182.20	31.59	4196.17
Columns Section IV	SHS 600 x 600 x 23.2	174.90	49.93	1336.29
Archway	SHS 400 x 400 x 7	145.61	138.91	9829.82
Truss Platform I	SHS 180 x 180 x 20	129.47	91.62	7636.20
Horizontal	SHS 800 x 800 x 24	186.23	42.67	1923.43

To determine more suitably sized members, the maximum axial stress experienced by each member type was recorded, and calculations using this stress value were carried out. The following analysis is for the primary column section at ground level, which approximates the cross-section of the beam as a square hollow section. Again, the following analysis is for the primary column section at ground level.

$$\text{Current load} = \phi N_s = 162.24 \times 95992 = 15574 \text{ kN} \quad (19)$$

$$\text{Suitable member area} = \frac{\phi N_s}{\phi f_y} = 83597 \text{ mm}^2 \quad (20)$$

To ensure the full area of the section is effective, i.e. the form factor k_f is equal to one, the following applies:

$$\frac{b}{t} \leq \frac{\lambda_{ey}}{\sqrt{\frac{f_y}{250}}} = 43.96 \quad (21)$$

For a square hollow section,

$$\text{member width} = \sqrt{\frac{b A_g}{4t}} = \sqrt{\frac{43.96 \times 83597}{4}} = 958.5 \text{ mm} \quad (22)$$

Checking section capacity,

$$N_c = \alpha_c N_s \leq N_s \quad (23)$$

$$\lambda_n = \left(\frac{l_e}{r} \right) \left(\sqrt{k_f} \right) \sqrt{\frac{f_y}{250}} = \left(\frac{7500}{382.5} \right) (\sqrt{1}) \sqrt{\frac{207}{250}} = 17.84 \quad (24)$$

$$\alpha_c = 0.989 \quad (25)$$

$$\phi N_c = 15109 \text{ kN} \quad (26)$$

$$\text{Maximum axial stress} = \frac{\phi N_c}{A_g} = \frac{15109}{83597} = 184.97 \text{ MPa} \quad (27)$$

Using this approach for all members, a static analysis was performed and the maximum axial stress for the optimised design was determined for all member types (see table 6).

TABLE 6. Proposed member sizes and capacities

Member	Suitable Members	Load Capacity (MPa)	Max Axial Stress (MPa)	Weight Savings (t)
Bracing Section I	EA 195.7 x 195.7 x 4.5	176.94	133.93	109.07
Columns Section I	SHS 958.5 x 958.5 x 21.8	184.97	155.73	110.30
Bracing Section II	EA 119.0 x 119.0 x 2.7	180.79	118.23	109.76
Columns Section II	SHS 573.0 x 573.0 x 13.0	163.93	119.29	276.64
Bracing Section III	EA 91.5 x 91.5 x 2.1	184.22	107.67	371.94
Columns Section III	SHS 415.3 x 415.3 x 9.4	158.68	108.66	437.30
Bracing Section IV	EA 87.9 x 87.9 x 2.9	185.53	121.69	210.65
Columns Section IV	SHS 405.3 x 405.3 x 9.2	165.60	125.47	400.72
Archway	SHS 300.3 x 300.3 x 6.8	183.59	146.43	216.17
Truss Platform I	SHS 150 x 150 x 15	115.10	109.73	215.25
Horizontal	SHS 433.0 x 433.0 x 9.9	184.20	82.32	808.53

All members satisfy the calculated axial load capacity. This has been carried out after only one iteration. Further iteration will reduce the size of members to their most optimum size for strength purposes, and is subject to further research. The size reductions thus far translate to a total weight saving of 3387 tonnes, which is 46% of the weight of the metal tower, or 34% of the weight of the entire structure including stairwells and elevator shafts. This can be further optimised after several iterations. It should be noted that the member capacities have been calculated using AS4100-1998, and thus are viewed as an approximation when calculating member capacities for wrought iron beams.

CONCLUSIONS

This finite element model of the Eiffel Tower has successfully enabled a thorough static and dynamic analysis of the structure under permanent, imposed, and wind loading, as well as under blast and earthquake loads due to an explosion. Dimensions have been justified through the use of original construction drawings, and are expected to largely replicate the existing tower. Nodes at the base are assumed to be fixed due to the rigid connection with the foundations, which themselves are very large and would require significant loading to affect. The tower utilises the full axial capacity of individual members by acting as a ‘truss of trusses’. As such, permanent and imposed loads are efficiently transferred to the primary columns through compression, while wind loads induce tensile forces in the windward legs and compressive forces in the leeward. Under blast loading, the tower experienced both ground vibrations and blast pressures. Ground vibrations induced a very small earthquake loading into the structure which was ignored in subsequent analyses. The blast pressure was significant, and a dynamic analysis of this revealed that further research is required into the damping qualities of the structure due to soil and mechanical properties. However, in understanding the worst case scenario, the blast was assumed to completely destroy several members in the adjacent leg. Under this weakened structure, it was observed that the tower would still be able to sustain static loads, at least for enough time for occupant evacuation. In final, an analysis of the structural members allowed an optimisation in the design of the tower by reducing the member cross sections while maintaining structural integrity.

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